

Math 141, F11

Quiz 2

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1.(4 points) Find a formula for the inverse of the function

$$f(x) = 4x^3 + 2.$$

Solution: $y = 4x^3 + 2$

$$y - 2 = 4x^3$$

$$x^3 = \frac{y-2}{4}$$

$$x = \sqrt[3]{\frac{y-2}{4}} \Rightarrow f^{-1}(x) = \sqrt[3]{\frac{x-2}{4}} \quad (\text{or } y = \sqrt[3]{\frac{x-2}{4}})$$

Problem 2.(6 points) Determine the following limits:

(a) $\lim_{x \rightarrow 1} \sqrt{2x^2 + x + 1}$

$$= \sqrt{2 \cdot 1^2 + 1 + 1} = \sqrt{4} = 2$$

(b) $\lim_{x \rightarrow \pi/3} \frac{1}{\cos x}$

$$= \frac{1}{\cos \frac{\pi}{3}} = \frac{1}{(\frac{1}{2})} = 2$$

(c) $\lim_{x \rightarrow 2^-} \frac{x+3}{x-2}$

As $x \rightarrow 2^-$, $x+3 \rightarrow 5$
 $x-2 \rightarrow 0$ and $x-2 < 0$

$$\Rightarrow \lim_{x \rightarrow 2^-} \frac{x+3}{x-2} = -\infty$$